Predict a position and a range

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Predict a future position and explain the limits of the prediction.

SUCCESS CRITERIA - I can...

1. Use matching units and include the starting position.
2. Report a central prediction and its bounds.
3. Explain how an uncertain measurement differs from a limit of the calculation.

Core: teacher selects from 1-9. **Stretch:** 10 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “initial” and “final” to explain your answer. Sentence starters are optional.

initial: at the beginning of the interval; subscript i

final: at the end of the interval; subscript f

bound: an upper or lower limit in a model

01 Warm-up: bring back an earlier idea

East is positive. Start at **6 m**, go to **-4 m**, then finish at **3 m**. **Sketch the route**; **calculate distance** and displacement. Distinguish both from final position.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $x_f = x_i + vt$; $\delta x_f = \delta x_i + t\,\delta v$.
Constant velocity; exact positive t and independent bounded starting position/velocity. δ marks a half-width, not a probability.

Worked reasoning: A cart starts at **1 m** and moves right at **2 m/s** for **3 s**. Central prediction: **1**+**2**×**3**=**7 m**. If its speed could be **1** to **3 m/s**, the endpoint range is **4** to **10 m**.

Your turn: A cart moves along a marked track.
1. **Show its start**, central predicted endpoint, and full possible endpoint range on the number line.
2. What must remain constant throughout the **3 s**?

Sentence starter: I predict … . I used … to make this prediction because … .



Position (m); right is positive

Marzano check - circle: ① ② ③

03 Units and a starting position

A train starts at x=**5 km** and moves at **+20 m/s** for **2 minutes**. **Calculate displacement** and final position in km.

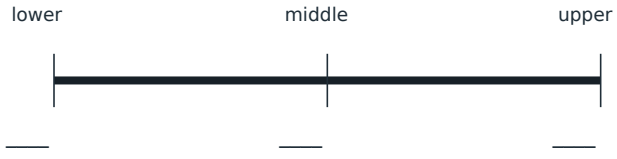
Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

04 Independent bounds

A motion sensor reports a cart's starting position as **10** ± **1 m** and its constant velocity as **2.0** ± **0.2 m/s**. Treat the two bounds independently; the elapsed time is exactly **5.0 s**.
1. **Find the central final position**, interval endpoints, half-width and full width.
2. **Label the diagram showing the** range of possible positions.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

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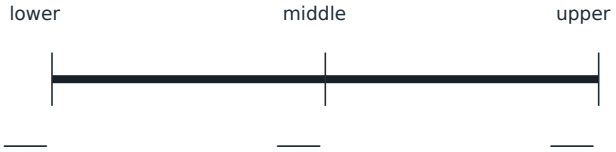
Continue down each column.

Sentence starters are optional. You may plan in any language.

05 A negative velocity

A robot starts at **30 ± 2 m** and moves with constant velocity **-3.0 ± 0.1 m/s**. The time is exactly **10 s**. Use the smallest and largest allowed inputs to **find its full final-position interval**.

Sentence starter: I found … [unit]. This value means … in this situation.

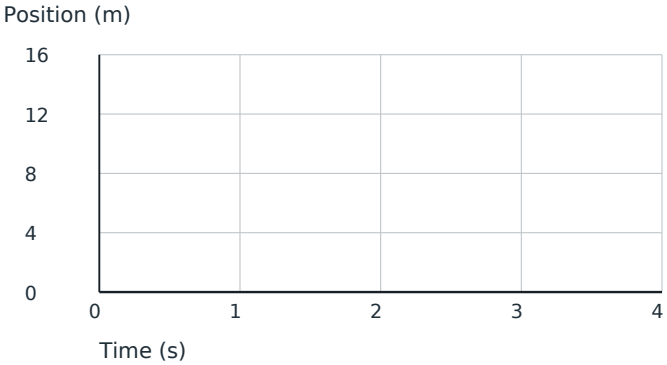


Marzano check - circle: ① ② ③

06 Show how the bound grows

For x_0=**4±1 m** and v=**2±0.5 m/s**, graph central, lower, and upper x(t) from **0** to **4 s**. **Label the endpoint range**.

Sentence starter: In my work, I used … to show … . I labeled it … .



Marzano check - circle: ① ② ③

07 Uncertain measurements or a problem with the prediction?

A precise speed measurement is used to **predict an asteroid position** with six months of constant velocity. **Explain why tiny measurement bounds** do not make the forecast exact.

Sentence starter: I predict … . I used … to make this prediction because … .

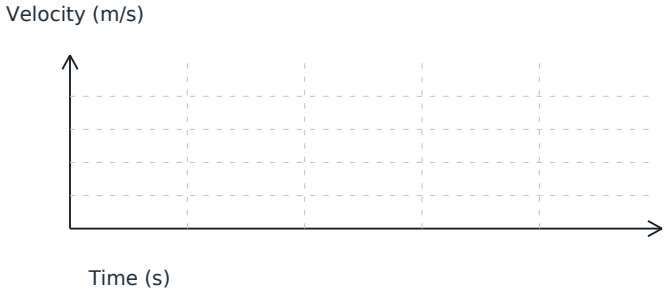
Marzano check - circle: ① ② ③

08 Review: connect an earlier idea

Velocity is **+3 m/s** for **5 s**, then **-2 m/s** for **3 s**. **Find displacement** and distance. **Explain the reference information needed** for an angle of **25°**.
Sketch the velocity–time graph; **label the given values**.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .



Marzano check - circle: ① ② ③

09 Exit check

x_0=**-5±0.5 m**, v=**3.0±0.1 m/s**, t=**10 s** exact.
1. **Give central final position**, its interval, and the full width.
2. **Name something you** are treating as true.

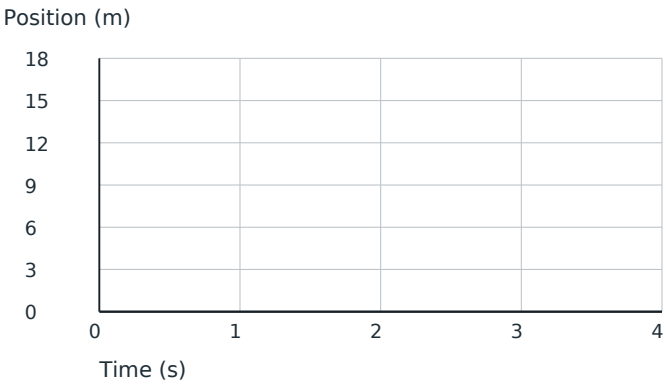
Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: ① ② ③

10 STRETCH: test the prediction

Initial position is **2±0.5 m**; constant velocity is **3±0.2 m/s**; time is exactly **4 s**. **Find the central prediction** and worst-case endpoint bounds.

Sentence starter: I predict … . I used … to make this prediction because … .



Marzano check - circle: ① ② ③